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Wang

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(45) **Date of Patent: Sep. 9, 2025**

(54) **CONTAINER ASSEMBLY**

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(72) Inventor: **Charlie Wang**, Chino Hills, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 594 days.

(21) Appl. No.: **17/746,876**

(22) Filed: **May 17, 2022**

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/378,326, filed on Jul. 16, 2021, now Pat. No. 11,896,050.

(60) Provisional application No. 63/190,568, filed on May 19, 2021, provisional application No. 63/084,118, filed on Sep. 28, 2020.

(51) **Int. Cl.**

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A24F 5/00	(2006.01)
A24F 7/00	(2006.01)
A24F 7/02	(2006.01)

(52) **U.S. Cl.**
CPC ***A24F 1/30*** (2013.01); ***A24F 5/00*** (2013.01); ***A24F 7/02*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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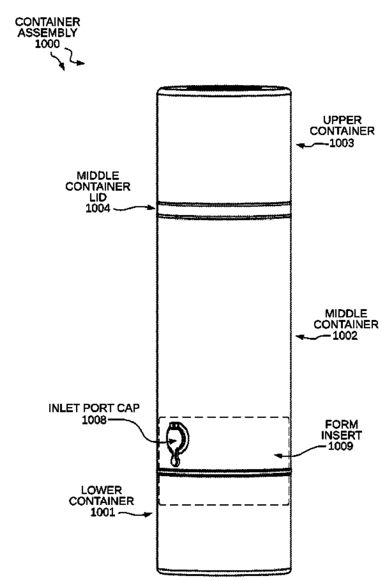
Primary Examiner — Dennis R Cordray
(74) *Attorney, Agent, or Firm* — Adibi IP Group, PC; Amir V. Adibi; Andrew C. Palmer

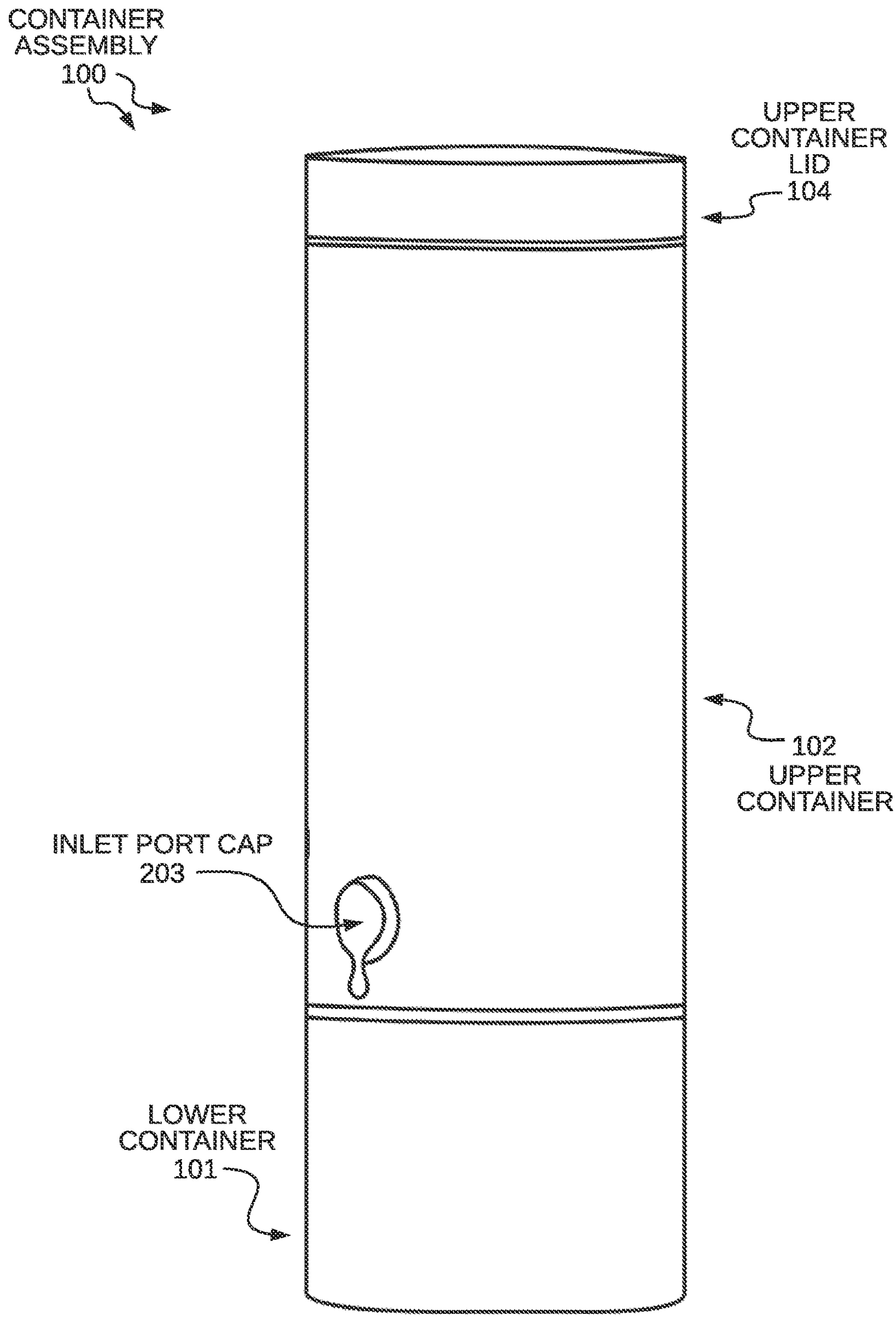
(57) **ABSTRACT**

A container assembly and method is provided for concealing and storing a water pipe. The container assembly, for example, is generally a cylindrical container comprising an upper container, a middle container, a middle container lid, a lower container, a percolator, an inlet port, a neck, and a bowl. The neck and the bowl are configured to be stored in the lower container. The upper container is configured to contain a grinder. The upper container and lower container are detachably attached to the middle container. The container assembly and method also involves sliding the percolator to and from an extracted position that permits a user to inhale combustible substances, to a concealed position within the middle container.

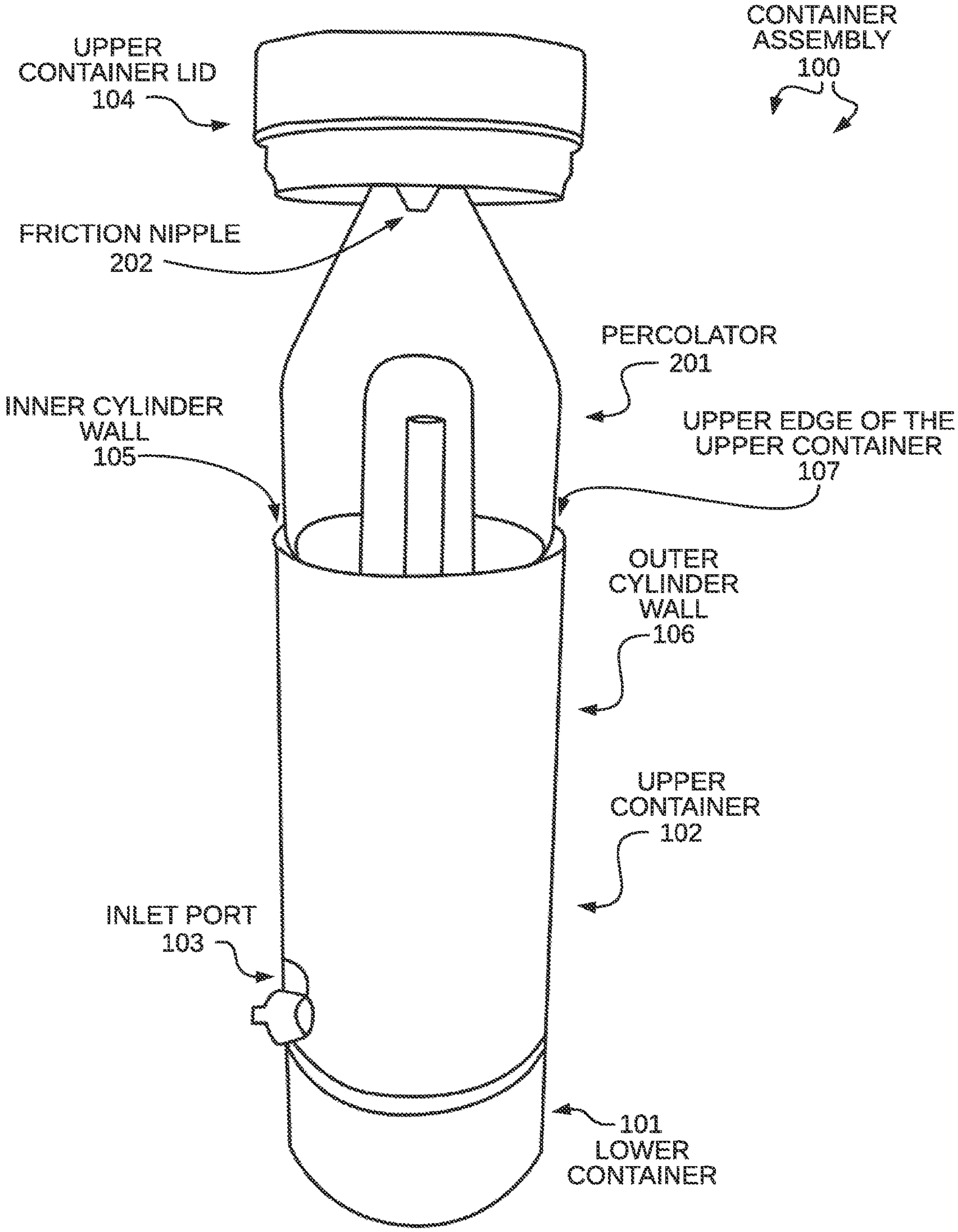
18 Claims, 17 Drawing Sheets



PERSPECTIVE VIEW OF ANOTHER EMBODIMENT

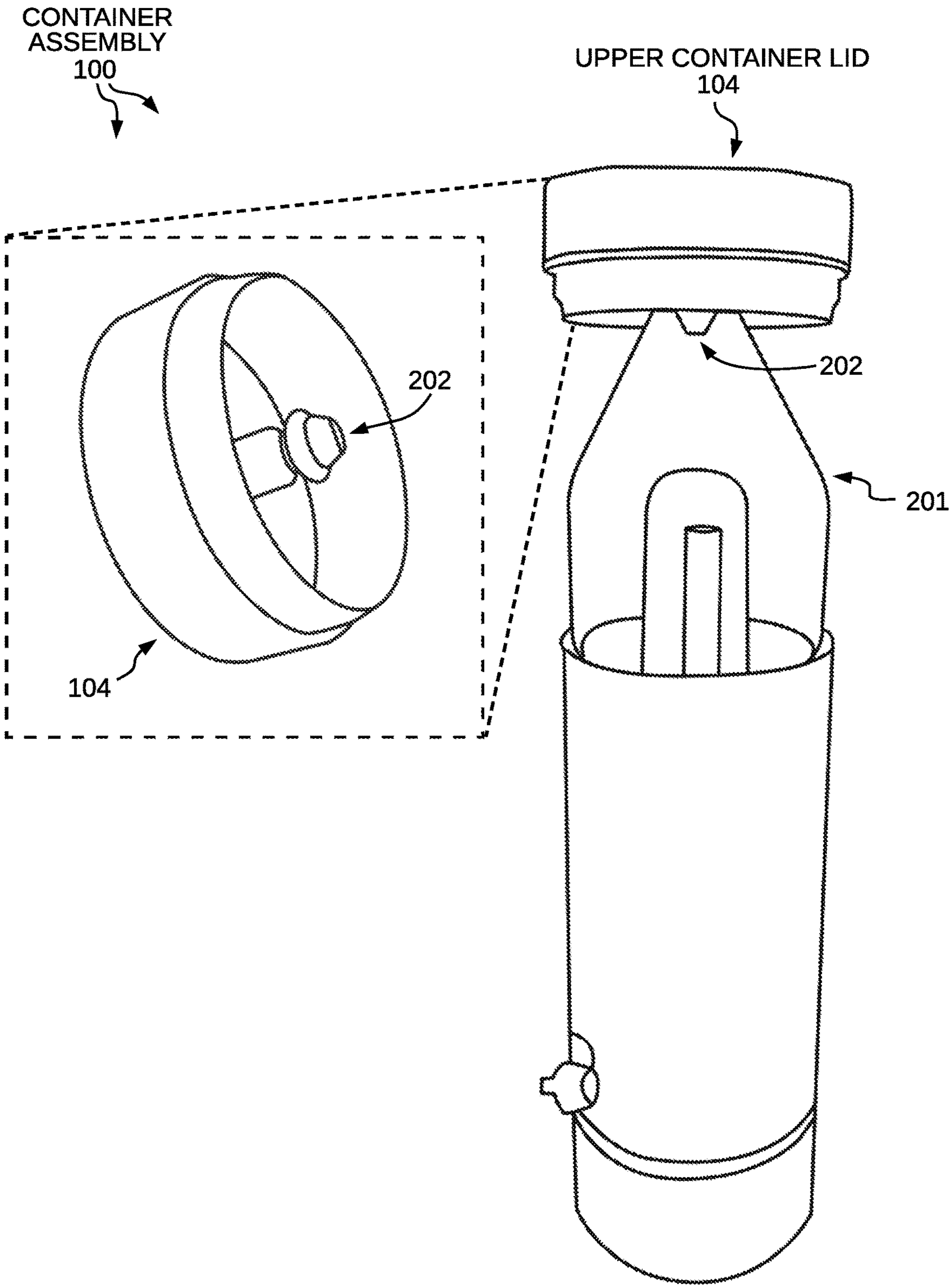


PERSPECTIVE VIEW OF CONTAINER ASSEMBLY
FIG. 1



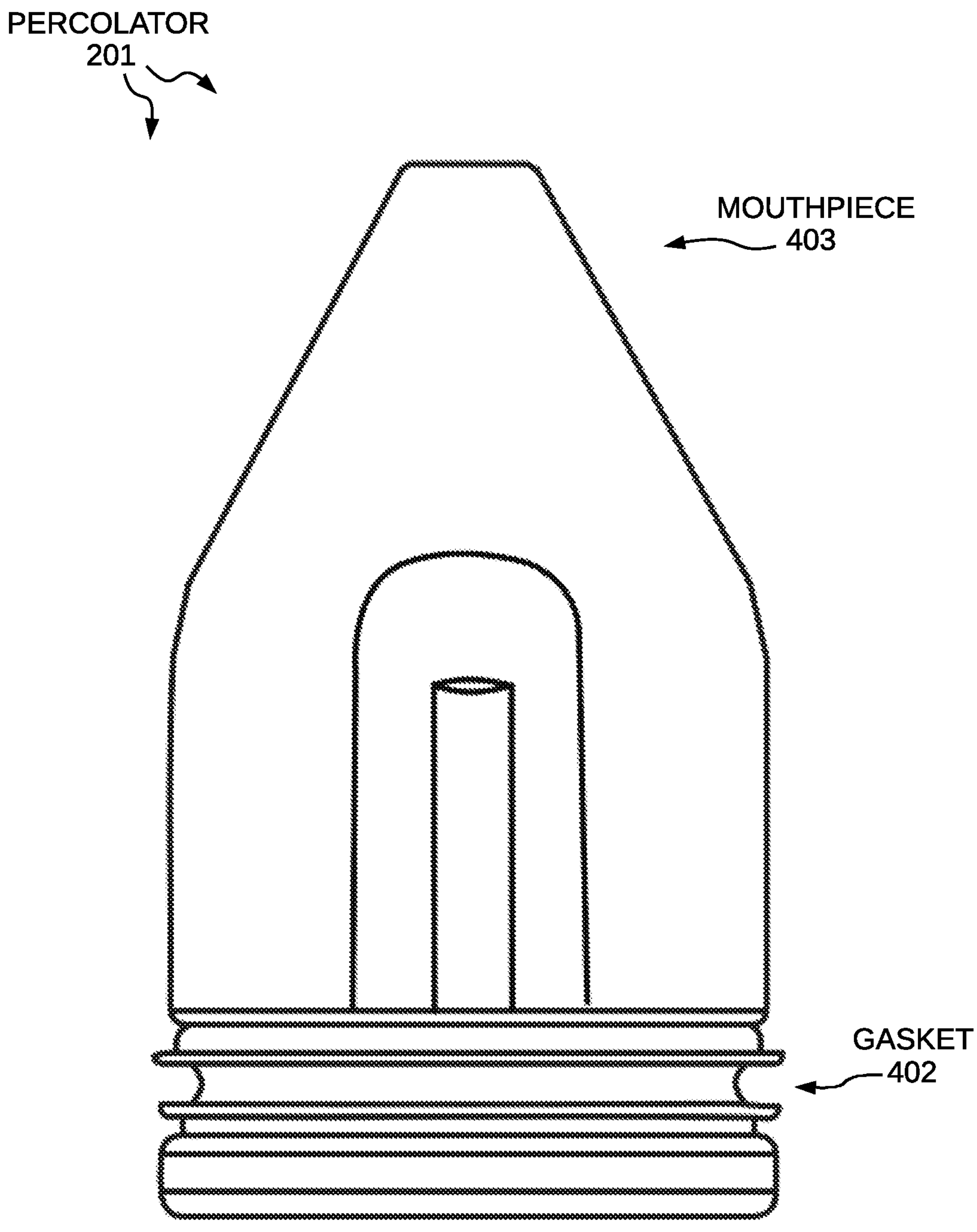
PERSPECTIVE VIEW OF UPPER CONTAINER LID AND PERCOLATOR

FIG. 2



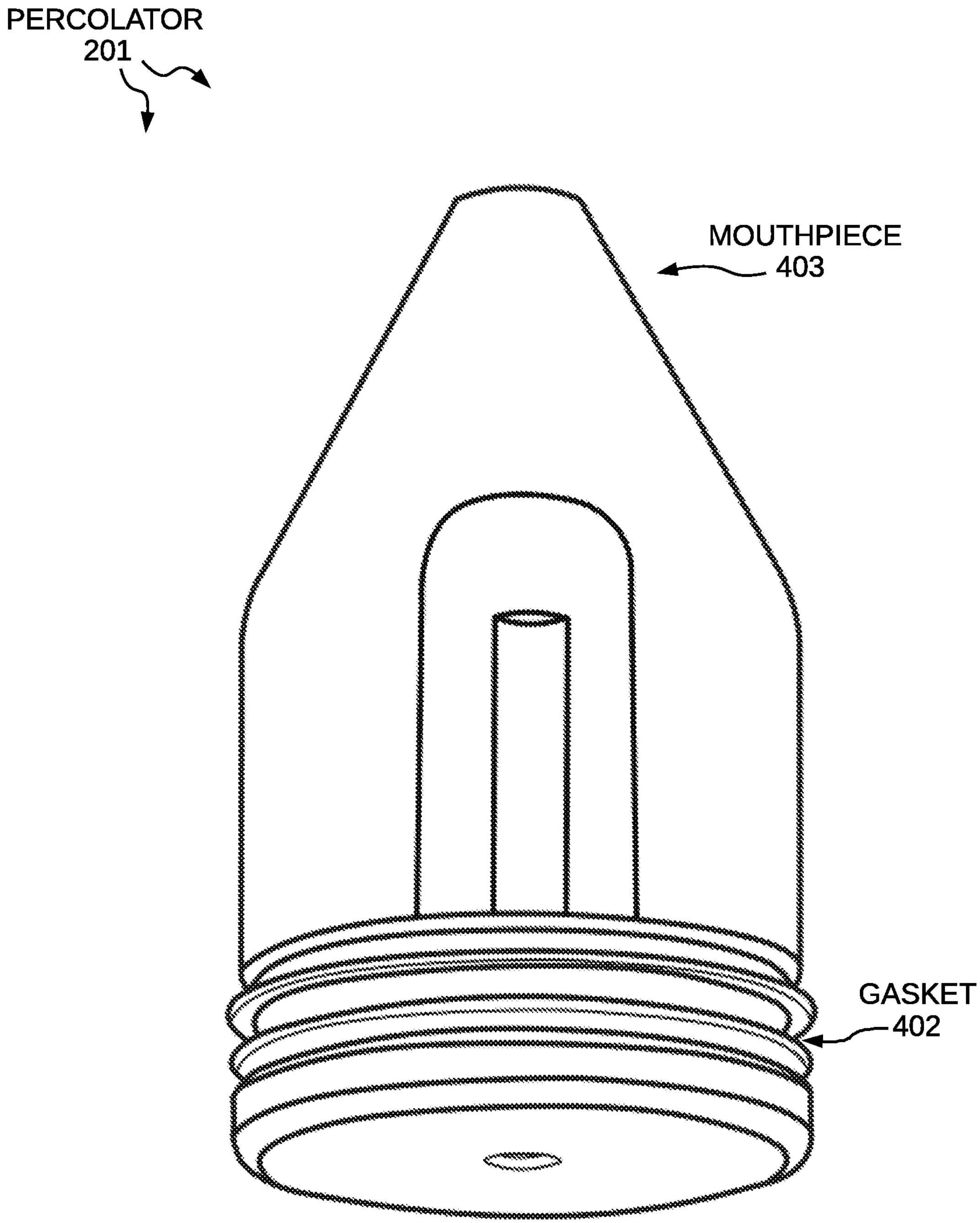
PERSPECTIVE VIEW OF UPPER CONTAINER LID AND PERCOLATOR

FIG. 3



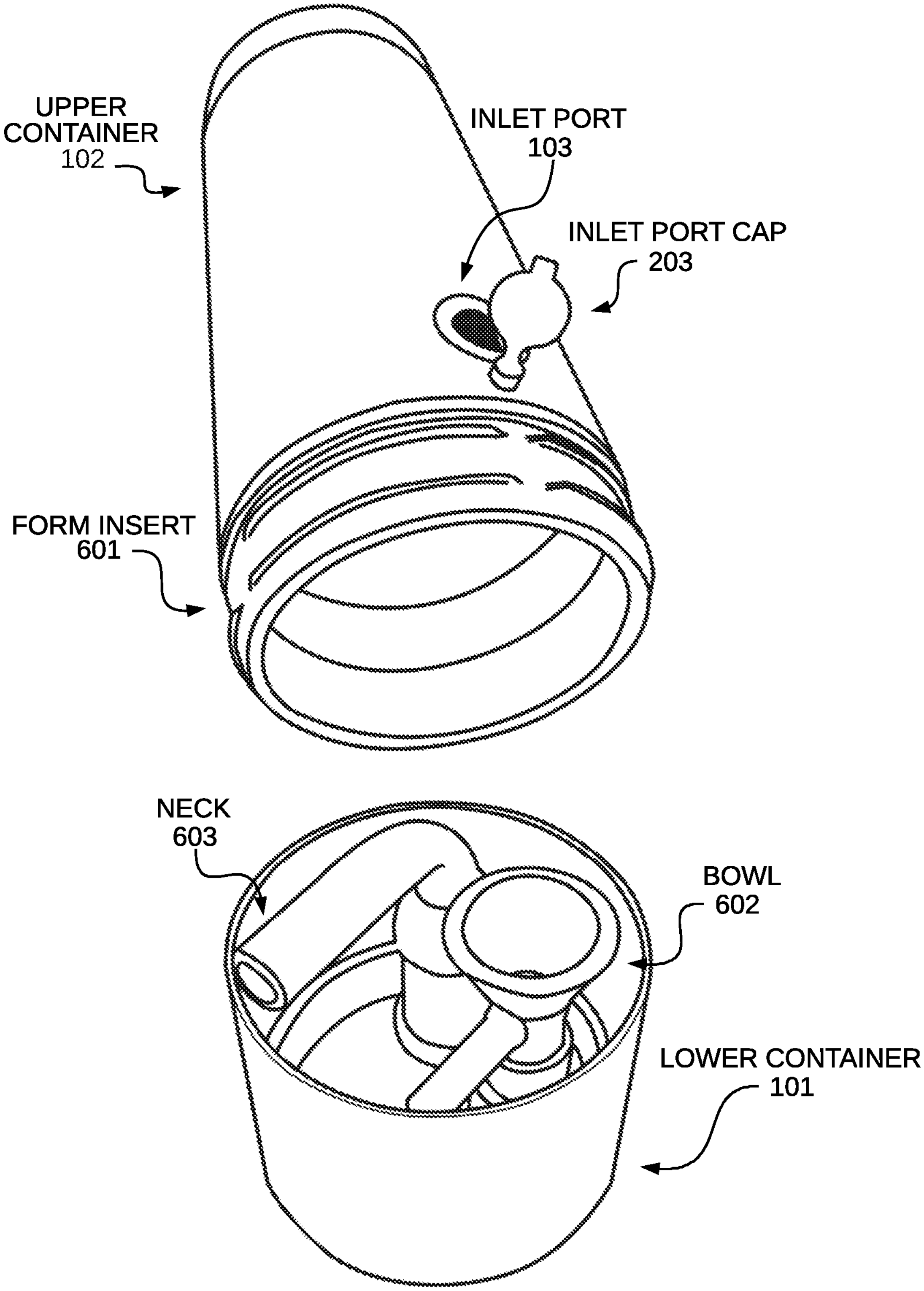
PERSPECTIVE VIEW OF PERCOLATOR

FIG. 4



PERSPECTIVE VIEW OF PERCOLATOR

FIG. 5



PERSPECTIVE VIEW OF LOWER CONTAINER

FIG. 6

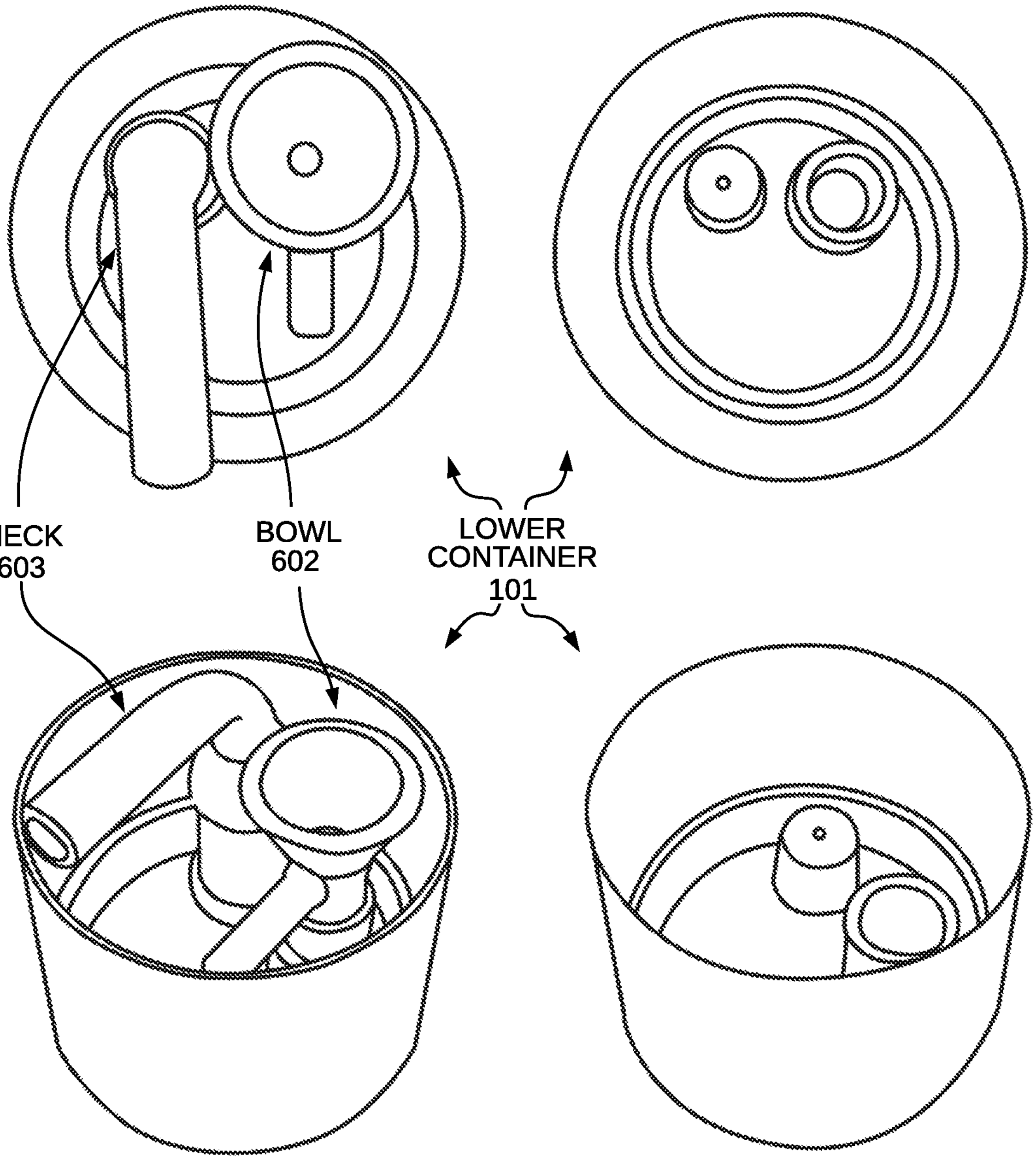


FIG. 7

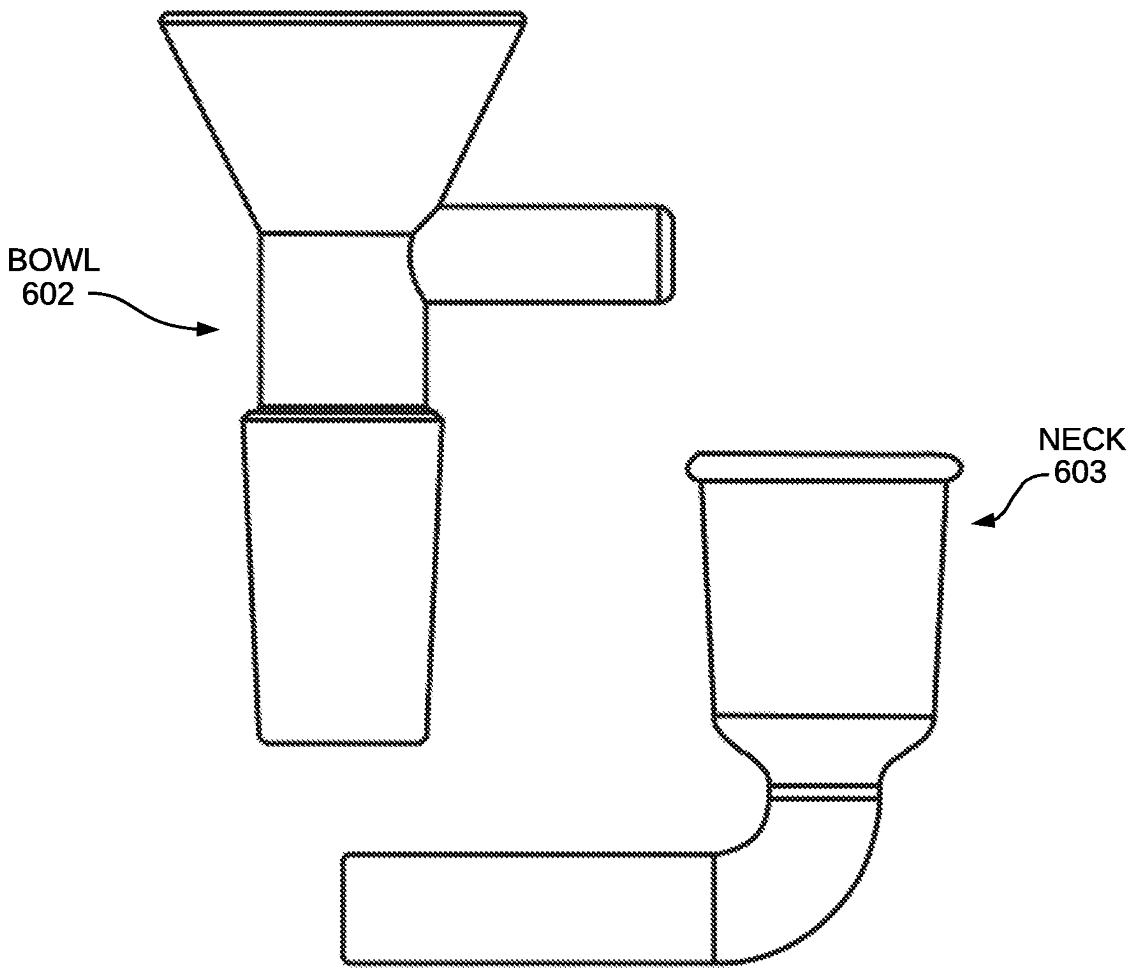


FIG. 8

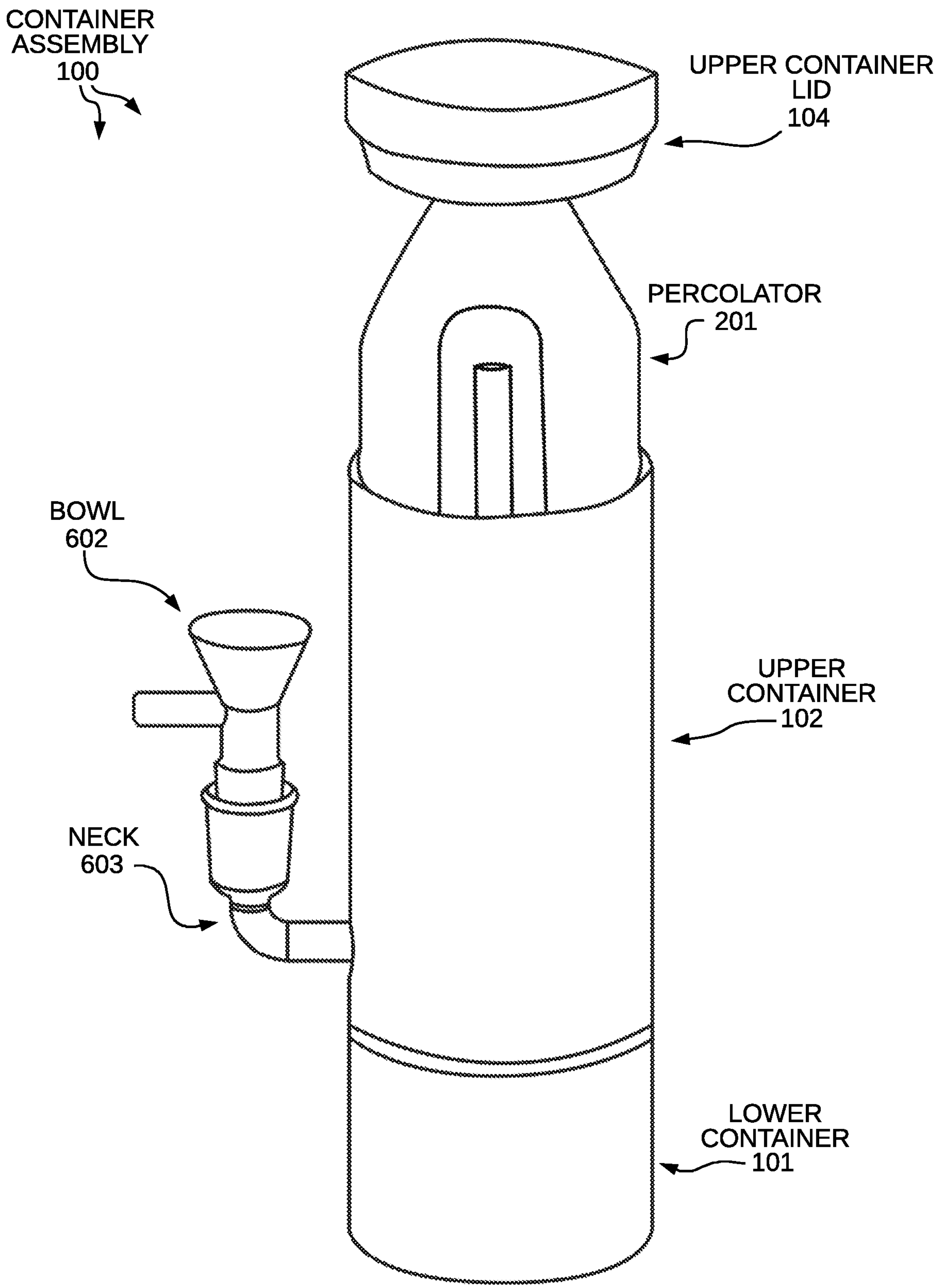
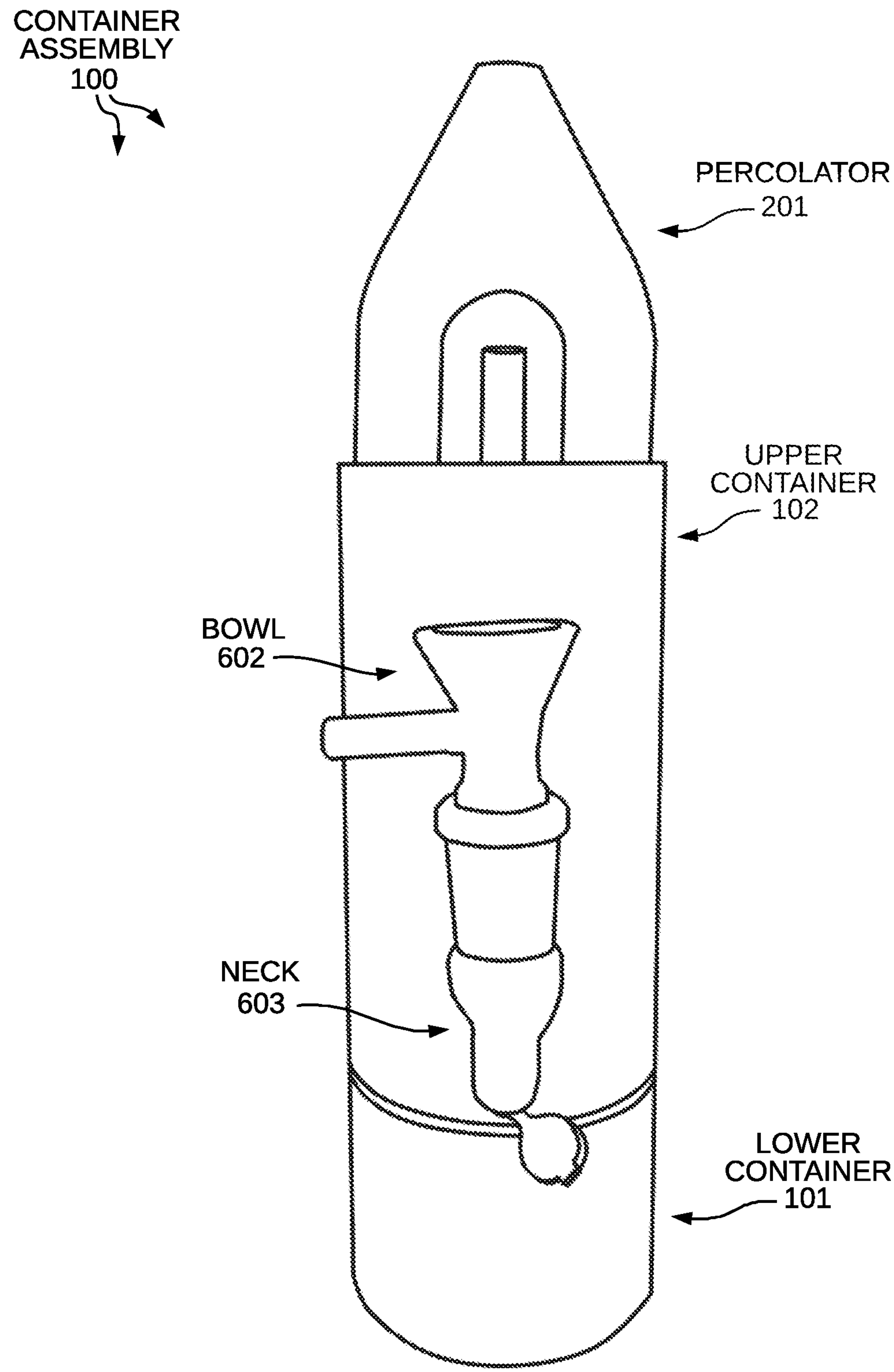
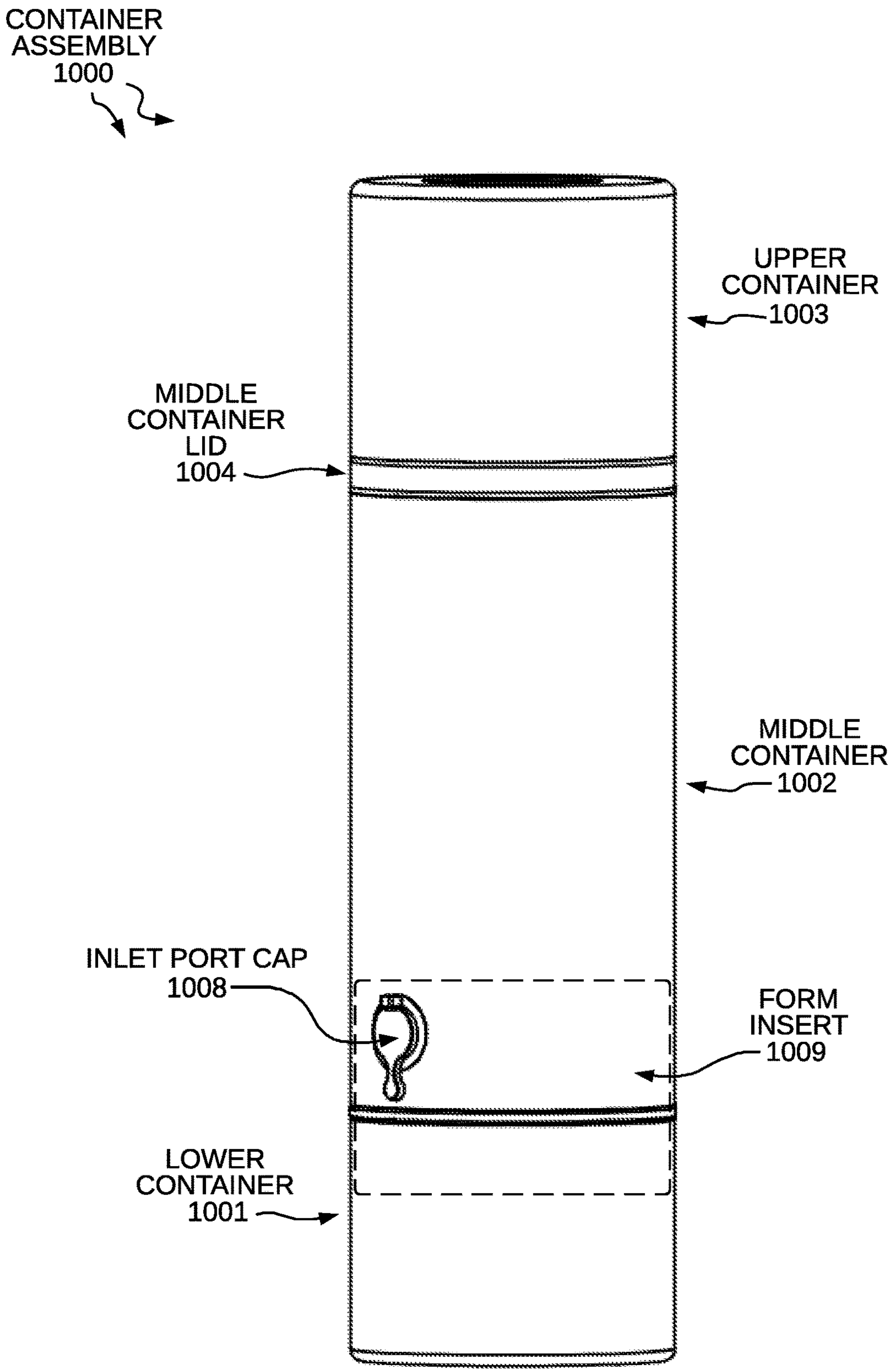

CONTAINER ASSEMBLY 100
UPPER CONTAINER LID 104
PERCOLATOR 201
BOWL 602
UPPER CONTAINER 102
NECK 603
LOWER CONTAINER 101

FIG. 9



PERSPECTIVE VIEW OF CONTAINER ASSEMBLY
FIG. 10



PERSPECTIVE VIEW OF ANOTHER EMBODIMENT

FIG. 11

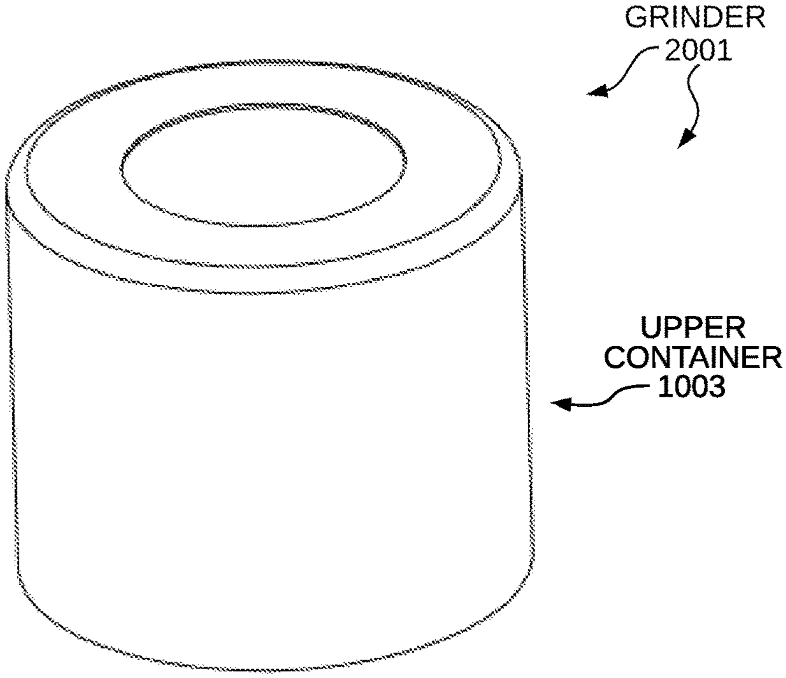

GRINDER
2001
UPPER
CONTAINER
1003

FIG. 12

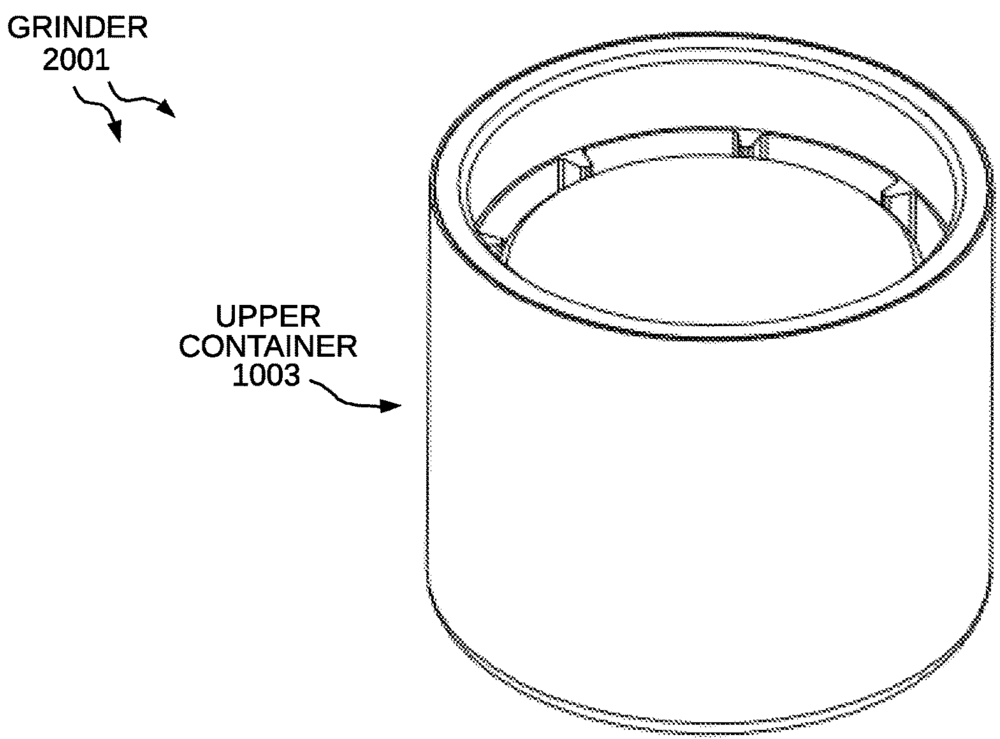
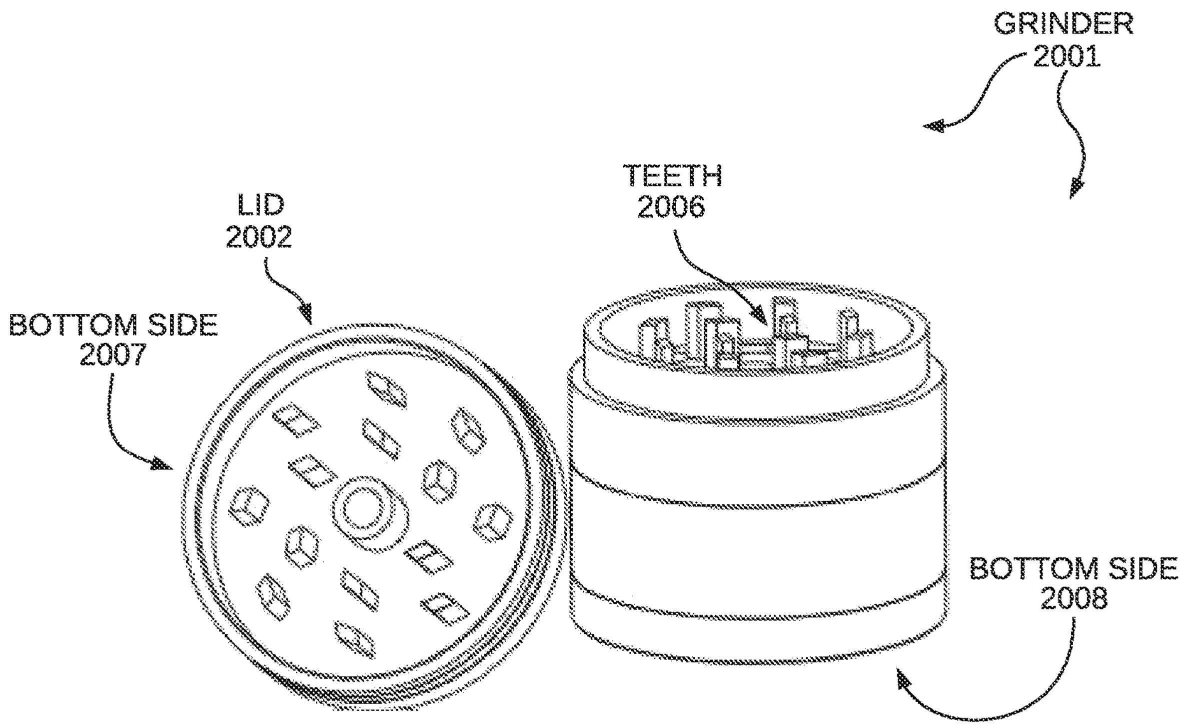
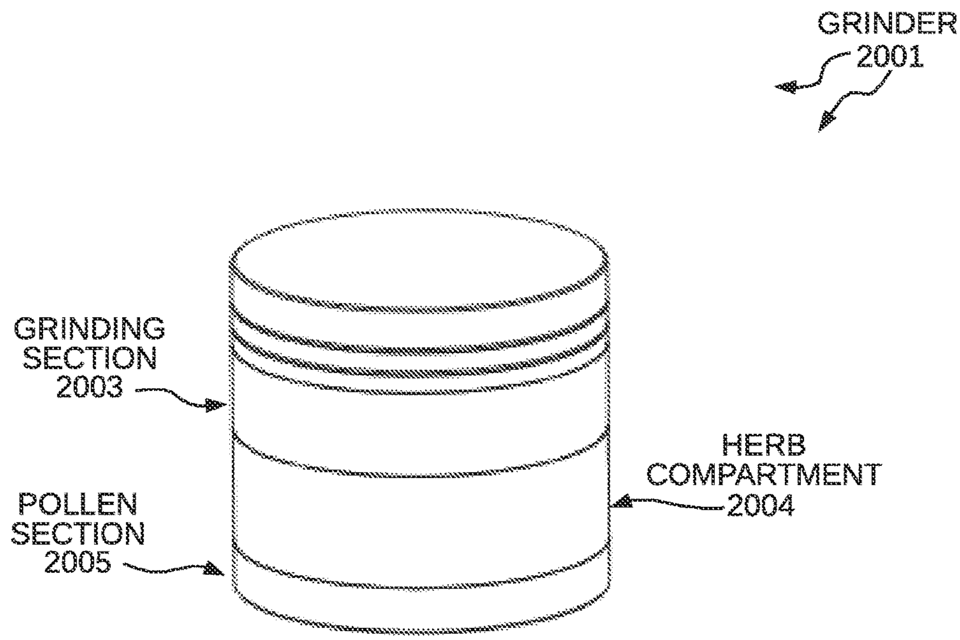

GRINDER
2001
UPPER
CONTAINER
1003

FIG. 13



PERSPECTIVE VIEW OF THE GRINDER

FIG. 14



PERSPECTIVE VIEW OF THE GRINDER

FIG. 15

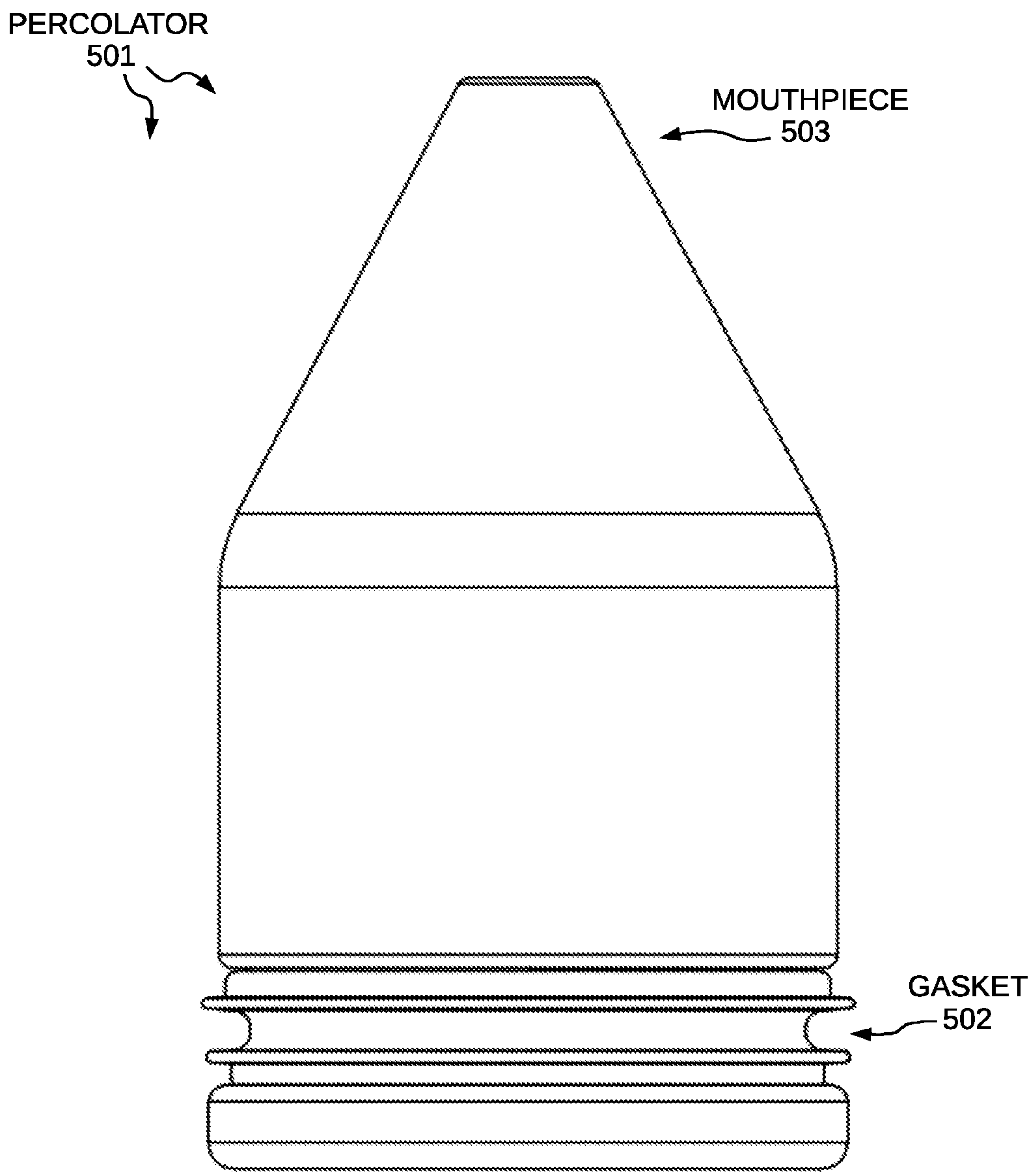
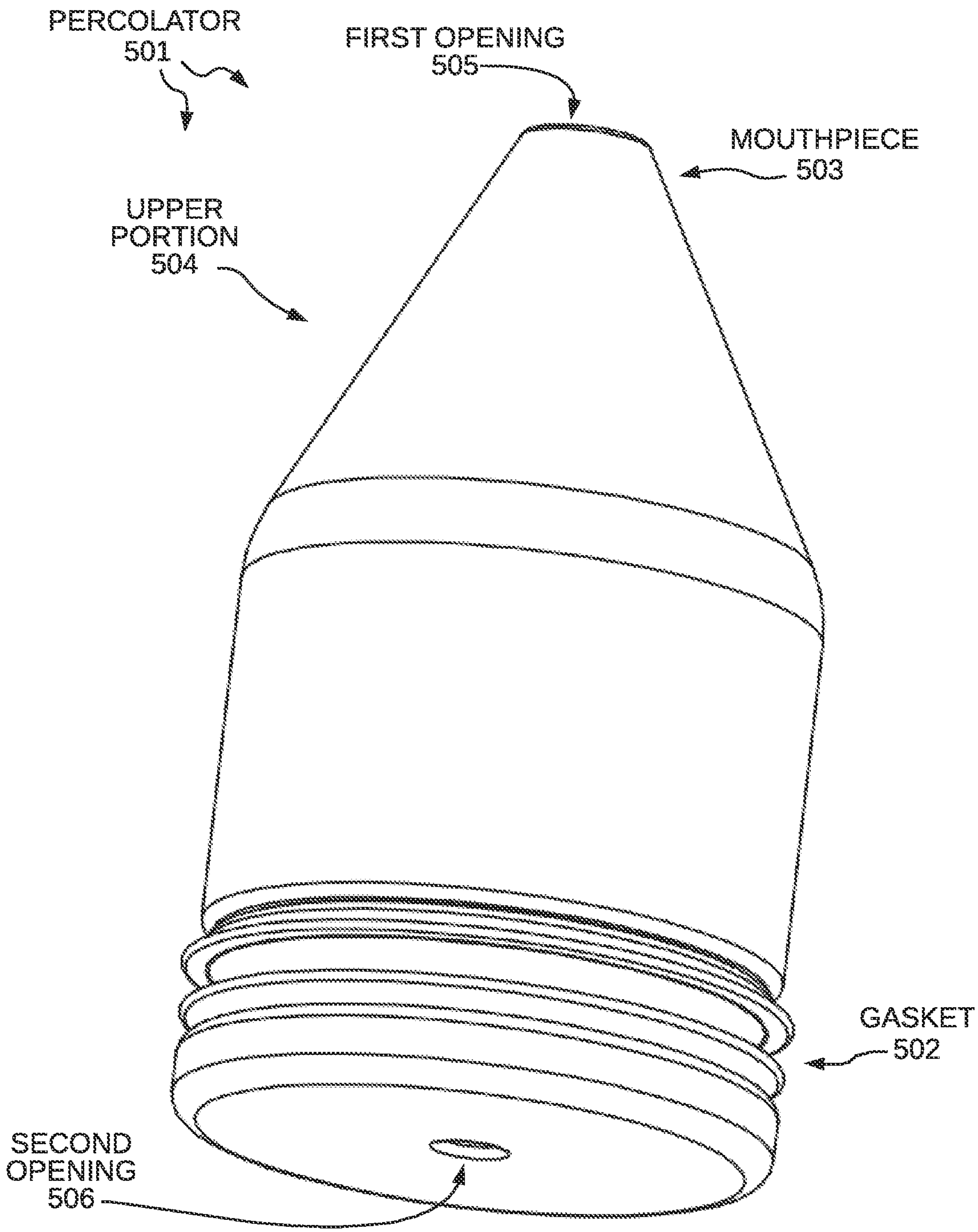


FIG. 16



PERSPECTIVE VIEW OF THE PERCOLATOR

FIG. 17

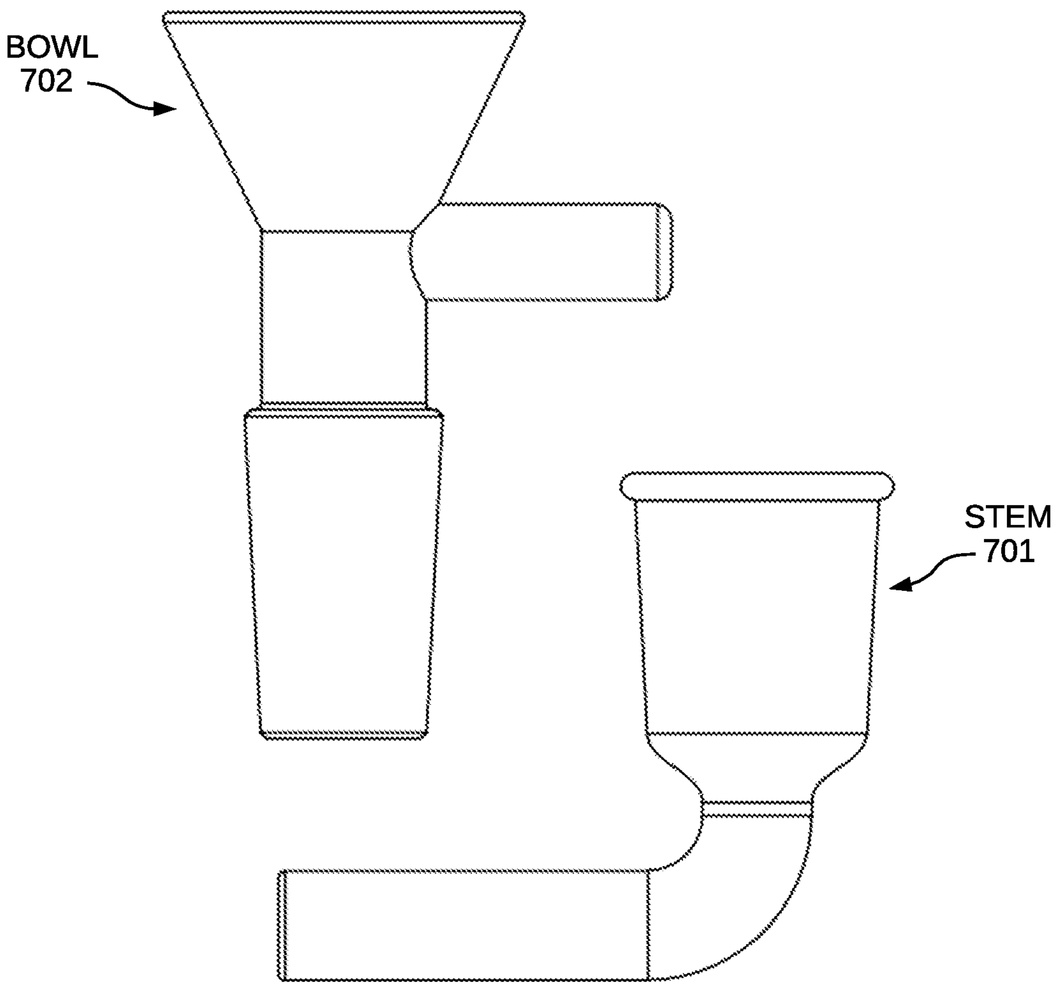
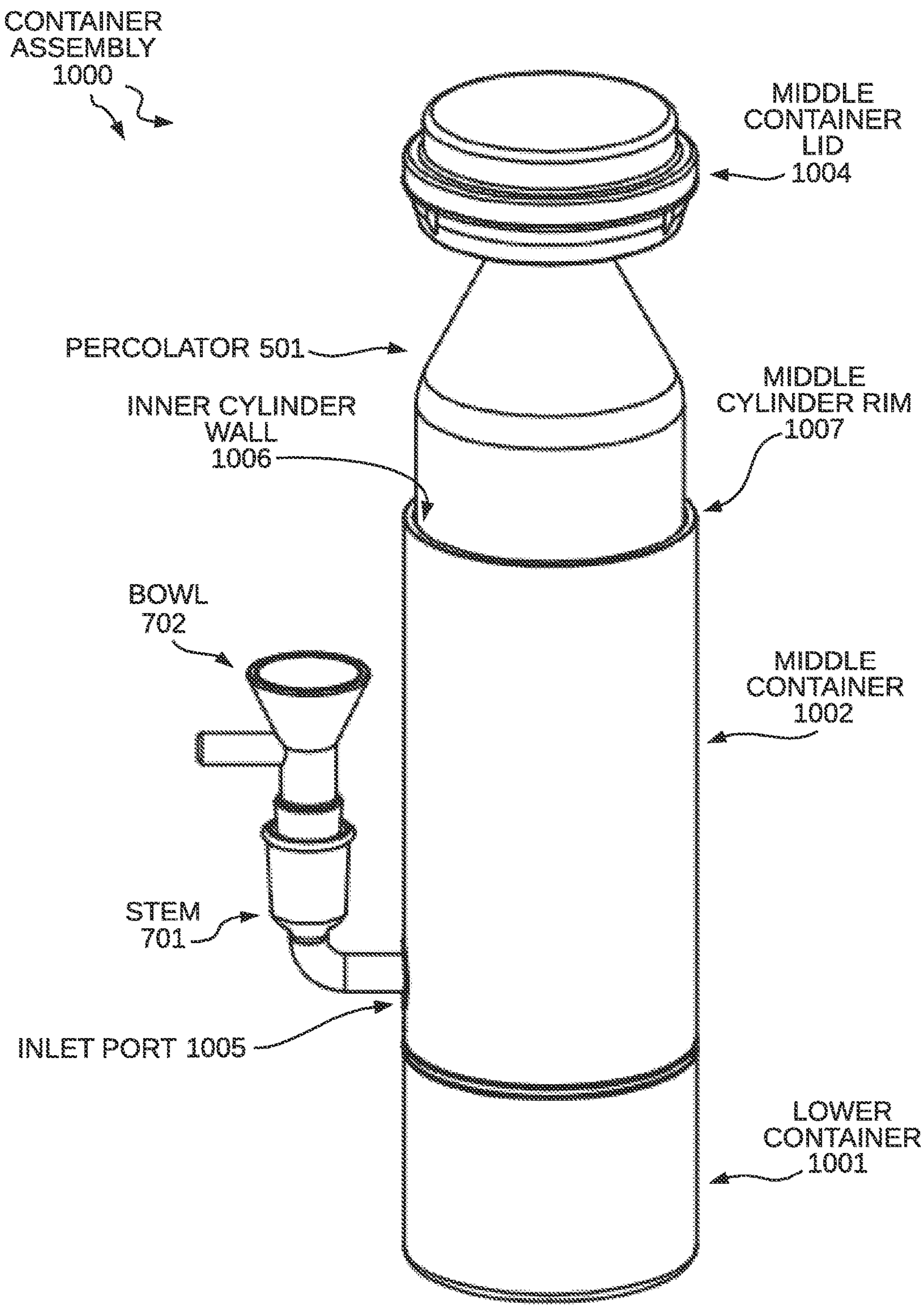


FIG. 18



PERSPECTIVE VIEW OF THE CONTAINER ASSEMBLY

FIG. 19

CONTAINER ASSEMBLY

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part that claims the benefit under 35 U.S.C. § 120 from U.S. nonprovisional patent application Ser. No. 17/378,326, entitled "Container Assembly," filed on Jul. 16, 2021. U.S. patent application Ser. No. 17/378,326 claims the benefit under 35 U.S.C. § 119 from U.S. provisional patent application Ser. No. 63/084,118, entitled "Container Assembly," filed on Sep. 28, 2020. This application also claims the benefit under 35 U.S.C. § 119 from U.S. provisional patent application Ser. No. 63/190,568, entitled "Container Assembly," filed on May 19, 2021. The subject matter of each of the foregoing documents is expressly incorporated herein by reference.

TECHNICAL FIELD

The present invention relates generally to an apparatus to store a smoking water pipe/bong and water pipe accessories such as a grinder; and more specifically, to a container assembly having a disguised appearance that permits a smoker to inhale combustible substances such as tobacco and medicinal herbs.

BACKGROUND INFORMATION

The use of water in a smoking apparatus is ancient, dating back at least several hundred years. An illustrative example of early smoking water pipes is Persian hookahs, wherein smoke from the substance being combusted is directed through a tube discharging below the surface of water in a chamber before passing therefrom through a second tube to the mouth of the user. More recently, various modern versions of the water pipes have been manufactured, sold, and used. These versions have not differed in the basic respect from the original hookah, comprising principally a chamber for holding the water, a bowl for the combustible substance communicating by a hollow tube to a point beneath the water, and an outlet tube from a smoke chamber formed above the surface of the water to the mouth of the user. These modern versions of the hookah have been produced in forms not adaptable to being carried by the user in a more compact, smaller, and portable version. Furthermore, the size and shape of these modern versions lack a disguised appearance and draws attention to the user when traveling with a water pipe in public. Since many users prefer to not draw attention to themselves, a desire exists for a container to disguise and store a smoking water pipe/bong and water pipe accessories, such as a grinder, in a portable container assembly.

SUMMARY

Prior to embodiments of the disclosed invention, there was no way to contain a water pipe and water pipe accessories in a discrete container assembly. Embodiments of the disclosed invention solve this problem and overcome the deficiencies of the prior art.

According to one embodiment, a container assembly is configured to store a water pipe. The container assembly comprises: an upper container, the upper container further comprising an inner cylinder wall, an outer cylinder wall, an upper edge, a lower edge, an upper container rim arranged at the upper edge of the upper container, and an inlet port,

wherein the inlet port extends through the outer cylinder wall and inner cylinder wall and into the upper container; an upper container lid, wherein the upper container lid is detachably coupled to the upper edge of the upper container; a lower container, wherein the lower container is detachably coupled to the lower edge of the upper container; a percolator, the percolator further comprising a gasket and a mouthpiece; a neck, wherein the neck is detachably coupled to the inlet port and in fluid communication with the percolator through the inlet port, wherein the neck is configured to be stored in the lower container; and a bowl, wherein the bowl is detachably coupled to the neck and in fluid communication with the neck and the percolator, wherein the bowl is configured to be stored in the lower container.

The present invention provides a compact and portable container assembly to conceal a water pipe and water pipe accessories that can be quickly assembled and disassembled.

According to another embodiment, a method for concealing a water pipe and water pipe accessories in a container assembly comprises: attaching an upper container lid to a mouthpiece arranged at an upper portion of a percolator, applying pressure to the upper container lid to slide the percolator from an extracted position to a concealed position, wherein the percolator is concealed within the upper container, attaching the upper container lid to the upper edge of the upper container, removing a lower container from the upper container, disconnecting a bowl from a neck, placing the bowl in the lower container, disconnecting the neck from an inlet port of the upper container, placing the neck in the lower container, and attaching the lower container to a lower edge of the upper container.

For various example embodiments, the following is applicable: an apparatus comprising means for performing a method of any of the claims.

Aspects of the present invention relate to an apparatus or device for a container assembly configured to store a water pipe or bong and various accessories for using the water pipe or bong and methods of using the container assembly.

These and other systems, methods, objects, features, and advantages of the present invention will be apparent to those skilled in the art from the following detailed description of the preferred embodiment and the drawings. All documents mentioned herein are hereby incorporated in their entirety by reference.

The foregoing, and other features and advantages of the invention, will be apparent from the following, more particularized description of the preferred embodiments of the invention, the accompanying drawings, and the claims.

Further details and embodiments and methods are described in the detailed description below. This summary does not purport to define the invention. The invention is defined by the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, where like numerals indicate like components, illustrate embodiments of the invention.

FIG. **1** shows a container assembly configured to store a water pipe according to one embodiment of the invention.

FIG. **2** shows an upper container lid and a percolator of the container assembly **100**.

FIG. **3** shows the upper container lid and percolator of the container assembly **100**.

FIG. **4** shows the percolator of the container assembly **100**.

FIG. **5** shows the percolator of the container assembly **100**.

FIG. **6** shows a lower container of the container assembly **100**.

FIG. **7** shows the lower container of the container assembly **100**.

FIG. **8** shows a neck and bowl of the container assembly **100**.

FIG. **9** shows a perspective view of the container assembly **100**.

FIG. **10** illustrates a perspective view of the container assembly **100**.

FIG. **11** illustrates a perspective view of another embodiment of an container assembly **1000** configured to store a water bong and water bong accessories.

FIG. **12** illustrates a perspective view of a grinder stored in an upper container of the container assembly **1000**.

FIG. **13** shows a perspective view of the grinder stored in the upper container of the container assembly **1000**.

FIG. **14** shows a perspective view of the grinder **2001** of the container assembly **1000**.

FIG. **15** shows a perspective view of the grinder **2001** of the container assembly **1000**.

FIG. **16** shows the percolator **501** of the container assembly **1000**.

FIG. **17** illustrates a perspective view of the percolator **501** of the container assembly **1000**.

FIG. **18** shows a stem and a bowl the container assembly **1000**.

FIG. **19** illustrates a perspective view of the container assembly **1000**.

DETAILED DESCRIPTION

Reference will now be made in detail to some embodiments of the invention, examples of which are illustrated in the accompanying drawings.

In various embodiments, a container assembly is disclosed having an innovative approach to conceal a water pipe or bong and its accessories, such as a grinder. The container assembly has a general cylindrical shape to disguise a water pipe or bong. The container assembly can be designed from a variety of materials to include steel, aluminum, copper, titanium, or plastic. The upper container and lower container can be colored the same or have different colors. The container assembly allows for quick concealment of the water pipe or bong when a user wishes to not draw attention to themselves. The components of the water pipe concealed within the container assembly can also be easily disassembled to allow for cleaning of the components of the water pipe.

FIG. **1** shows a perspective view of a container assembly **100** in accordance with one embodiment. The container assembly **100** is comprised of an inner cylinder wall **105**, an outer cylinder wall **106**, an upper container **102**, an upper container lid **104**, an inlet port **103** (displayed with an inlet port cap **203** attached to the inlet port), and a lower container **101**, wherein the upper container **102** and the lower container **101** are detachably coupled to each other and capable of being segregated from each other. By way of example, one embodiment of the container assembly **100** is configured to store a water pipe or bong and permits the user to quickly assemble or disassemble the water pipe concealed within the upper container **102** and the lower container **101**. An upper container rim is arranged at an upper edge **107** of the upper container **102**. A lower container **101** is detachably coupled to a lower edge of the upper container **102**.

The material of container assembly **100** is one selected from a group including but not limited to: steel, aluminum, copper, titanium, plastic, or a combination thereof. The preferred material is aluminum because it is the most cost-efficient, is easy to clean, and offers protection to the components of the water pipe and water pipe accessories.

FIGS. **2** and **3** show a perspective view of the upper container lid **104** and percolator **201**. The upper container lid **104** is detachably coupled to the upper edge **107** of the upper container **102** when the percolator **201** is completely inserted into the upper container. This position permits the percolator **201** to be concealed within the upper container **102**. In another embodiment, the upper container lid **104** comprises a friction nipple **202** that when inserted into the percolator **201** (as displayed in FIG. **2**), allows the percolator **201** to be extracted from its concealed position within the upper container **102**. Once the percolator **201** is fully extracted (as displayed in FIG. **2**), the percolator **201** is in a position that permits a smoker to inhale combustible substances, such as tobacco and medicinal herbs, through the percolator **201**.

FIGS. **4** and **5** show a perspective view of the percolator **201**. In a one embodiment, the percolator **201** includes a gasket **402** and a mouthpiece **403**. The mouthpiece **403** is arranged at an upper portion of the percolator **201**. The percolator **201** is arranged within the upper container **102** and slides along the inner cylinder wall **105** to the upper edge **107** of the upper container **102**. To extract the percolator **201** and the mouthpiece **403**, the upper container lid **104** is detached from the upper container **102** and the inlet port cap **203** is detached from the inlet port **103**, and with the friction nipple **202** fully inserted into the mouthpiece **403**, the upper container lid **104** is pulled upward in a vertical motion from the upper container **102** to extract the percolator **201** from a concealed position within the upper container **102**. When the percolator **201** is pulled from the concealed position inside of the upper container **102** to an extracted position, the gasket **402** is pressed against the upper container rim to form a watertight seal. The percolator **201** is one selected from a group including but not limited to: dome, inline, circular, showerhead, honeycomb, tree, matrix, swiss, donut, turbine, or a combination thereof. The preferred percolator **201** is a dome percolator.

FIGS. **6** and **7** show a perspective view of the lower container **101**. In one embodiment, the lower container **101** comprises a form insert **601** that fits the shape of the upper container **102** and the lower container **101**. The form insert **601** is detachably coupled to the lower edge of the upper container **102** and the lower container **101** is detachably coupled to the form insert. In another embodiment, a neck **603** and a bowl **602** can be attached to the form insert **601** to prevent the neck **603** and the bowl **602** from cracking, chipping, or breaking while stored inside the lower container **101**.

FIG. **8** shows a perspective view of the neck **603** and the bowl **602**. The neck **603** is in fluid communication with the percolator **201** when a user connects the neck **603** to an inlet port **103** extending through the outer cylinder wall and inner cylinder wall and into the upper container **102**. The inlet port **103** includes an inlet port cap **203** to protect the inlet port **103** and prevent the inside of the upper container **102** from getting contaminated. The neck **603** may be configured to be stored in the lower container **101**. When the lower container **101** is detached from the upper container **102**, a user may then access the lower container **101** to remove the neck **603**. The bowl **602**, when attached to the neck **603**, is in fluid communication with the neck **603** and the percolator **201**.

The bowl **602** may be configured to be stored in the lower container **101**. The neck **603** and bowl **602** are designed from a variety of materials. The preferred material of the neck **603** and bowl **602** is glass.

FIGS. **9** and **10** show a perspective view of the container assembly **100** wherein the percolator **201** is in an extracted position, the upper container lid **104** is detached from the percolator **201**, the neck **603** is attached to the inlet port **103**, and the bowl **602** is attached to the neck **603**. The container assembly **100** can now be used as a traditional water pipe or bong.

FIG. **11** is a diagram of a container assembly **1000** in accordance with another embodiment. The container assembly **1000** is comprised of an inner cylinder wall, an outer cylinder wall, an upper container **1003**, a middle container **1002** having an inlet port **1005** and inlet port cap **1008**, a middle container lid **1004**, and a lower container **1001**, wherein the upper container **1003**, the middle container **1002**, and the lower container **1001** are detachably coupled to each other and capable of being segregated from each other. By way of example, one embodiment of the container assembly **1000** is configured to store a water bong or pipe and permits the user to quickly assemble or disassemble the water bong concealed within the upper container **1003**, the middle container **1002**, and the lower container **1001**. A middle container rim is arranged at an upper edge of the middle container **1002**. The lower container **1001** comprises a form insert **1009**. The form insert is detachably coupled to the lower edge of the middle container **1002**, and the lower container is detachably coupled to the form insert.

An upper container **1003** is detachably coupled to the middle container lid **1004**. The middle container lid **1004** is detachably coupled to the upper edge of the middle container **1002**.

The material of container assembly **1000** is one selected from a group including but not limited to: steel, aluminum, copper, titanium, plastic, or a combination thereof. The preferred material is aluminum because it is the most cost-efficient, easy to clean, and offers protection to the components of the water bong and water bong accessories.

FIGS. **12** and **13** show a perspective view of the grinder **2001** stored in the upper container **1003** of the container assembly **1000**. The upper container **1003** comprises a push release to detach and remove the grinder **2001** from the upper container **1003**. Preferably, the grinder **2001** is capable of removing unwanted stems and/or seeds from combustible substances such as tobacco and medicinal herbs.

FIGS. **14** and **15** show a perspective view of the grinder **2001** in one configuration. In one embodiment, the grinder **2001** comprises a lid **2002**, a grinding section **2003**, which includes a plurality of teeth **2006**, wherein the plurality of teeth **2006** are disposed of in a pattern on a bottom side **2007** of the lid **2002** and a bottom side **2008** of the grinding section **2003**. In another embodiment, the grinder **2001** includes an herb compartment **2004** to collect the tobacco and/or herb that has been grinded to a finer material. In yet another embodiment, the grinder **2001** comprises a pollen section **2005** which includes a screen to sift the pollen from the tobacco and/or herb. In yet another embodiment, the grinder **2001** is smaller and comprises only a grinding section **2003** to permit the user to store other water bong accessories, such as a lighter, in the upper container **1003**.

FIGS. **16** and **17** show a perspective view of the percolator **501**. The percolator **501** comprises a gasket **502** and a mouthpiece **503**. The percolator **501** has an upper portion **504** and a first opening **505** opposite the second opening **506**. The mouthpiece **503** is arranged at an upper portion **504** of the percolator **501**. The percolator **501** is arranged within the middle container **1002** and slides along the inner cylinder wall **1006** to the middle container rim **1007**. To access the percolator **501** and extract the mouthpiece **503**, a user may simply detach the middle container lid **1004** from the middle container **1002** and pull the middle container lid **1004** upward in a vertical motion from the middle container **1002**. When a user pulls the percolator **501** from a concealed position inside of the middle container **1002** to an extracted position, the gasket **502** is pressed against the middle container rim to form a watertight seal. The percolator **501** is one selected from a group including but not limited to: dome, inline, circular, showerhead, honeycomb, tree, matrix, swiss, donut, turbine, or a combination thereof. The preferred percolator **501** is a dome percolator.

FIG. **18** shows a perspective view of the stem **701** and bowl **702**. Stem **701** may also be referred to as a "neck". The stem **701** is in fluid communication with the percolator **501** when a user connects the stem **701** to an inlet port **1005** extending through the outer cylinder wall and inner cylinder wall and into the middle container **1002**. The stem **701** may be configured to be stored in the lower container **1001**. A user can remove the lower container **1001** and then access the lower container **1001** to remove the stem **701**. The bowl **702**, when attached to the stem **701**, is in fluid communication with the stem **701** and the percolator **501**. The bowl **702** may be configured to be stored in the lower container **1001**. The stem **701** and bowl **702** are designed from a variety of materials. The preferred material of the stem **701** and bowl **702** is glass.

FIG. **19** shows a perspective view of the container assembly **1000** wherein the percolator **501** is in an extracted position and the stem **701** is attached to the inlet port **1005** and bowl **702** is attached to the stem **701**. The container assembly **1000** can now be used as a traditional water bong or pipe.

Although certain specific embodiments are described above for instructional purposes, the teachings of this patent document have general applicability and are not limited to the specific embodiments described above. Accordingly, various modifications, adaptations, and combinations of various features of the described embodiments can be practiced without departing from the scope of the invention as set forth in the claims.

What is claimed is:

1. A container assembly comprising:

an upper container;

a middle container, wherein the middle container includes an inner cylinder wall, an outer cylinder wall, an upper edge, a lower edge, a middle container rim arranged at the upper edge of the middle container, and an inlet port, and wherein the inlet port extends through the outer cylinder wall and inner cylinder wall and into the middle container;

a lower container, wherein the lower container is detachably coupled to the lower edge of the middle container;

a percolator, wherein the percolator includes a gasket and a mouthpiece;

a neck, wherein the neck is detachably coupled to the inlet port and is in fluid communication with the percolator through the inlet port, and wherein the neck is configured to be stored in the lower container; and

a bowl, wherein the bowl is detachably coupled to the neck and is in fluid communication with the neck and the percolator, and wherein the bowl is configured to be stored in the lower container.

2. The container assembly of claim **1**, wherein the percolator is arranged within the middle container and configured to slide along the inner cylinder wall to the middle container rim.

3. The container assembly of claim **2**, wherein the gasket is pressed between the percolator and the middle container rim to form a watertight seal.

4. The container assembly of claim **1**, wherein the mouthpiece is arranged at an upper portion of the percolator.

5. The container assembly of claim **1**, wherein the middle container lid is detachably coupled to the mouthpiece.

6. The container assembly of claim **5**, the middle container lid further comprising:

a friction nipple, wherein the friction nipple is detachably coupled to the mouth piece and is configured to be inserted into the mouthpiece of the percolator.

7. The container assembly of claim **1**, the inlet port further comprising:

an inlet port cap.

8. The container assembly of claim **1**, wherein the lower container is detachably coupled to the lower edge of the middle container via a form insert.

9. The container assembly of claim **8**, wherein the form insert is adapted to secure the neck and the bowl when the neck and the bowl are stored in the lower container.

10. A method for disassembling and storing a water pipe in a container assembly, comprising:

attaching a middle container lid of the container assembly to a mouthpiece of the water pipe arranged at an upper portion of a percolator of the water pipe;

applying pressure to the middle container lid to slide the percolator from an extracted position to a concealed position, wherein the percolator is concealed within a middle container of the container assembly;

disconnecting a bowl of the water pipe from a neck of the water pipe;

placing the bowl in a lower container of the container assembly;

disconnecting the neck from an inlet port of the middle container; and

placing the neck in the lower container.

11. The method of claim **10**, further comprising:

securing the middle container lid to an upper edge of the middle container; and

securing the lower container to a lower edge of the middle container.

12. The method of claim **10**, wherein attaching the middle container lid to the mouthpiece further comprises inserting a friction nipple into a mouthpiece of the percolator.

13. The method of claim **10**, wherein placing the bowl in the lower container further comprises securing the bowl to a form insert, and wherein the form insert is detachably coupled to the middle container.

14. The method of claim **13**, wherein placing the neck in the lower container further comprises securing the neck to the form insert.

15. The container assembly of claim **1**, further comprising:

a middle container lid, wherein the middle container lid is detachably coupled to the upper edge of the middle container.

16. The container assembly of claim **15**, wherein the upper container is detachably coupled to the middle container lid.

17. The container assembly of claim **1**, further comprising:

an upper container lid, wherein the upper container lid is detachably coupled to an upper edge of the upper container.

18. The container assembly of claim **1**, further comprising:

a grinder, wherein the grinder includes a lid, a grinding section, a plurality of teeth and a bottom side.

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